Constraint-Based Signal Formation
Structural Conditions of Signal Output Derivation Using Dimensional Human Field Theory (DHFT)

DHFT Projection Series — Information Systems | Paper II

Cara Tonucci
Independent Researcher
May 4, 2026 — Version 1
Cara.Tonucci@gmail.com
ORCID: https://orcid.org/0009-0003-9048-5821

Abstract

Information systems operate under load.

Signal formation is determined by structural relations of Load, Capacity, and Boundary.

This paper defines structural conditions for signal formation using Dimensional Human Field Theory (DHFT), a constraint-based system defined by invariant variables: Load (*L*), Capacity (*C*), and Boundary (*B*).

Signal outputs are derived from system conditions and do not constitute system variables.

Signals do not correspond to instantaneous system state.

No additional variables are introduced.

This paper constitutes a constraint-preserving projection of Dimensional Human Field Theory (DHFT) onto information systems.

I. Scope and Position

This paper defines structural conditions for signal formation under load using DHFT.

Projection applies to:

- individual systems
- organizational systems
- institutional systems
- artificial systems

All claims are restricted to structural relations of Load (*L*), Capacity (*C*), and Boundary (*B*).

Projection does not modify system definition.

No additional variables are introduced.

This paper does not:

- define meaning or message constructs
- interpret communication or signal content
- evaluate informational accuracy
- prescribe communication frameworks or models

II. System Operation

An information system operates under load.

The system:

- receives load
- distributes load across interacting systems
- regulates load through boundary conditions

System conditions are determined by structural relations of Load (*L*), Capacity (*C*), and Boundary (*B*).

Signal outputs are derived from these system conditions and do not constitute system variables.

III. Structural Specification

Load (*L*)
Structural demand within and across systems.

Capacity (*C*)
System limit for absorbing and processing load.

Boundary (*B*)
Structural condition governing admission, containment, and transfer of load across systems.

Signal outputs do not introduce additional variables and remain dependent on these invariant variables.

IV. Structural Relations

Structural relations determine system conditions.

Load:

- applies to a system
- transfers across systems
- accumulates across systems

Capacity:

- limits system response
- establishes thresholds for stability

Boundary conditions:

- regulate transfer
- constrain propagation

All system conditions are determined by relations between Load (*L*), Capacity (*C*), and Boundary (*B*).

Signal outputs are derived from these relations.

V. System State Conditions

A system produces stable signal conditions when:

- load remains within capacity
- boundary conditions regulate transfer

A system produces unstable signal conditions when:

- load exceeds capacity
- boundary conditions fail

Failure is sustained instability across interacting systems.

Signal outputs reflect system conditions across load, capacity, and boundary relations and do not correspond to instantaneous system state.

VI. Constraint Preservation

All admissible signal descriptions preserve:

- Load (L), Capacity (C), and Boundary (B)
- structural relations
- constraint conditions

Projection preserves:

- invariant variables
- structural relations
- admissibility conditions
- dependency ordering

Signal outputs are derived outputs and do not define system structure.

No substitution of variables is admissible.

No additional variables are admissible.

All admissible descriptions reduce to these constraints.

VII. Non-Equivalence

This system is not equivalent to:

- communication models
- message transmission systems
- semantic or linguistic frameworks
- interpretive or cognitive processing models

Signal outputs are not equivalent to messages, meaning, or content constructs.

Substitution of structural signal conditions with domain-specific constructs is non-admissible.

Similarity in terminology or outcome does not establish equivalence.

Partial overlap does not constitute substitution.

Structural equivalence with external systems is non-admissible.

DHFT defines a constraint-based system.

DHFT is not reducible to external models.

VIII. Boundary of Claims

This paper defines structural conditions for signal formation under load.

This paper does not:

- define meaning or message content
- interpret communication or signals
- evaluate informational accuracy
- prescribe communication frameworks or systems

Projection does not redefine signal formation at the domain level.

All claims are restricted to structural relations of Load (*L*), Capacity (*C*), and Boundary (*B*).

IX. Canonical System Statement

Dimensional Human Field Theory (DHFT) is a closed explanatory system defined by invariant variables:
Load (*L*), Capacity (*C*), Boundary (*B*).

All admissible system descriptions preserve these variables and structural relations.

Canonical system definition resides exclusively in DHFT Technical Series publications.

Applications operate under DHFT constraints and do not constitute system definition.

X. Canonical Series Statement

The DHFT Projection Series preserves invariant variables, structural relations, and constraint conditions.

Across DHFT projection domains, outputs are not independent phenomena. Outputs are structural expressions under constraint.

No projection introduces additional variables or redefines system mechanics.

The Information Systems series includes:

- Paper I — structural definition of information systems
- Paper II — signal formation under constraint
- Paper III — signal attribution and load assignment
- Paper IV — information stability and integrity

These documents define a dependency-ordered projection of Dimensional Human Field Theory (DHFT) onto information systems.

Each paper extends prior structural conditions without modifying invariant variables or system relations.

All projections operate under DHFT constraints and remain reducible to canonical system structure.

XI. System Reference

The system is specified in:

Dimensional Human Field Theory (DHFT): Architectural Identification of a Closed Explanatory System

DOI: 10.5281/zenodo.19075948

All admissible system descriptions reduce to Load (*L*), Capacity (*C*), and Boundary (*B*).

No additional variables are introduced at the minimal layer.

This document is part of the DHFT Projection Series — Information Systems.

XII. Copyright